

Distinction Reorganization and the Geometry of Understanding

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Abstract

Scientific revolutions, memory reconsolidation, archive rematching, representation learning, mechanistic interpretability, and ontology repair are customarily treated as distinct problems belonging to different disciplines. This paper argues that they are manifestations of a single underlying process: the reorganization of admissible distinctions into a structure with greater reachability and recoverability. The central claim is that understanding is not the accumulation of descriptions but a *trajectory* of distinction reorganization: a sequence of structures $\mathcal{D}_0 \rightarrow \mathcal{D}_1 \rightarrow \dots$ in which each transition expands reachability, reduces residual density, and retroactively remaps the accumulated observational archive. We introduce a formal framework — distinction geometry — in which distinctions are organized as dependency graphs, reachability is defined via recovery operators, and residual structure density $\bar{\Phi}$ is given both an ideal formulation (via Kolmogorov complexity) and a computable operational proxy (via distance to the learned admissibility manifold). The framework is used to formalize Kuhnian paradigm change as a quantitative dynamical process (revolution = $\Delta\bar{\Phi} \ll 0$), to state the World-Model Crossover and Directionality Principle as the formal boundary between observation-centered and model-centered epistemology, to introduce the Representational Revision Principle as a guard against conflating epistemological and ontological revision, and to prove the Observational–Interventional Separation Theorem as the formal foundation for the interpretability reframing. The framework unifies six independent research programs — Frozen Processes, Persistent Anomalies, Waves of Collapse, Admissibility Geometry, Distinguishability Geometry, and RSVP — as cross-sections of a single geometric object.

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1 Introduction

There is a question that sits uncomfortably beneath almost every contemporary discussion of artificial intelligence, cognitive science, and the philosophy of science, yet is rarely stated directly:

What is the difference between a system that predicts correctly and a system that understands?

The standard answer — that understanding is reliable, general, or robust prediction — is demonstrably insufficient. A system may achieve high predictive accuracy on every benchmark while collapsing exactly the distinctions that would allow it to intervene, explain, or generalize beyond its training distribution. This is not a failure of capability. It is a failure of faithfulness: the representation has discarded structure that matters.

This paper proposes a different answer. Understanding arises when a representation preserves and reorganizes the *admissible distinctions* of a domain — the distinctions whose collapse would impair navigation, intervention, or explanation relative to the purposes for which the representation exists. Understanding advances not gradually but through *collapse events*: discontinuous reorganizations of the distinction structure that expand reachability, enable rematching of prior observations, and increase the domain’s navigability.

This claim unifies a cluster of apparently unrelated phenomena. Scientific revolutions are collapse events at civilizational scale. Memory reconsolidation is collapse at the scale of individual belief. Archive rematching is collapse in the domain of stored records. Ontology repair in AI systems is collapse in the domain of learned representations. Mechanistic interpretability is, properly understood, the project of verifying whether collapse has occurred and what structure it has revealed.

The framework rests on three core concepts.

Admissible distinctions are those whose collapse would impair performance on a given task or family of tasks. Admissibility is always relative to a purpose; the same difference in the world may be admissible for one task and inadmissible for another.

Reachability is the capacity to express, recover, and operate on a distinction within a given representation. A distinction that exists in the world but is not reachable from a representation does not belong to that representation’s understanding.

Ontological deficit is the frozen residue of collapsed admissible distinctions: the structured anomalies that a representation cannot compress, the persistent noise that resists current vocabulary, the pressure that accumulates before a collapse event releases it.

Together these concepts define a geometry — the geometry of understanding — in which progress is not a smooth accumulation but a sequence of reorganizations, each of which expands the space of what is reachable.

The paper is structured as follows. Section 2 establishes that every nontrivial representation destroys distinctions (the Projection Theorem) and introduces the core

vocabulary. Section 3 defines admissibility and ontological deficit and distinguishes them from predictive failure. Section 4 introduces residual structure density Φ as the quantitative measure of accumulated distinction failure. Section 5 reinterprets Kuhnian paradigm change as distinction reorganization, making the sociology of science geometrically precise. Section 6 states the World-Model Crossover Principle and the Directionality Principle as the formal boundary between observation-centered and model-centered epistemology. Section 7 shows that human memory behaves exactly as the framework predicts. Section 8 applies the framework to AI interpretability. Section 9 shows how the framework unifies several independent research programs. Section 10 states the master claim and its corollaries.

2 Projection and Distinction Destruction

2.1 Representations as projections

Every representation is a map from a richer domain to a poorer one. This is not a defect but a necessity: a map that preserved every detail of the territory would be useless. The question is not whether information is lost but which information.

Definition 2.1 (Representation and projection). Let Ω be a domain of cases. A *representation* is a surjective map $\pi : \Omega \rightarrow M$ where M is the representational space. We call π a *projection* and M the *model space*.

Definition 2.2 (Distinction). A *distinction* on Ω is a binary relation $d \subseteq \Omega \times \Omega$ such that $d(x, y)$ holds when x and y are treated differently for some purpose. A distinction induces a partition of Ω into equivalence classes under the complementary relation $\neg d$.

A flat set of such relations is too weak a substrate for the framework. The concepts of strata, ceilings, and reachability all presuppose that distinctions stand in dependency relations to one another: collapsing a high-level distinction may destroy downstream ones without affecting peers. We therefore organize distinctions into a richer structure.

Definition 2.3 (Distinction dependency graph). A *distinction dependency graph* $\mathcal{G}_D = (V, E)$ over domain Ω is a directed acyclic graph in which:

- (i) Each vertex $v \in V$ corresponds to a distinction d_v on Ω .
- (ii) A directed edge $(u \rightarrow v) \in E$ indicates that d_v *depends on* d_u : drawing d_v presupposes that d_u has already been drawn. Collapsing d_u therefore propagates upward, destroying all d_v reachable from u via directed paths.

The *distinction structure* $\mathcal{D}$ is the pair $(\Omega, \mathcal{G}_D)$.

Example 2.4 (Biological taxonomy as a distinction graph). Consider the distinctions: *living thing* (d_1), *animal* (d_2), *vertebrate* (d_3), *mammal* (d_4), *dog* (d_5). The dependency graph has edges $d_1 \rightarrow d_2 \rightarrow d_3 \rightarrow d_4 \rightarrow d_5$. Collapsing d_4 (the mammal distinction) propagates upward and destroys d_5 , but leaves d_1, d_2, d_3 intact. This asymmetry —

which the flat binary-relation definition cannot represent — is what gives distinction structures their geometry.

Definition 2.5 (Graph reachability and collapse). A distinction d_v is *collapsed* by π if there exist $x, y \in \Omega$ with $d_v(x, y)$ but $\pi(x) = \pi(y)$.

A distinction d_v is *graph-reachable* in $\mathcal{G}_D$ from a set $S \subseteq V$ if there exists a directed path from some $u \in S$ to v in $\mathcal{G}_D$. Collapsing a distinction d_u renders graph-unreachable all distinctions depending on d_u .

The following result is elementary but foundational.

Theorem 2.6 (Projection Theorem). *Let $\pi : \Omega \rightarrow M$ be any nontrivial projection (i.e., $|M| < |\Omega|$). Then π collapses at least one distinction on Ω .*

Proof. Since $|M| < |\Omega|$, by the pigeonhole principle there exist distinct $x, y \in \Omega$ with $\pi(x) = \pi(y)$. The pair (x, y) constitutes a collapsed distinction. $\square$

The Projection Theorem is not a theorem about compression in the technical sense; it is a theorem about representation in general. Every simplification, abstraction, categorization, and model necessarily collapses some distinctions. The question of epistemology is which distinctions are collapsed and which are preserved.

2.2 The Weierstrass obstruction

A tacit assumption in many representational frameworks is that a sufficiently rich finite vocabulary can approximate any signal to arbitrary precision. This assumption fails for signals with persistent fine-scale structure.

Remark 2.7 (Weierstrass obstruction). The Weierstrass function $f(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$ (with $0 < a < 1$, b odd, $ab > 1 + \frac{3\pi}{2}$) is continuous everywhere and differentiable nowhere. No finite library of smooth templates achieves finite residual on such a signal: new oscillations appear at every scale of magnification.

For a representation whose template library contains only smooth primitives, the Weierstrass-type signal constitutes a permanent ontological deficit. The deficit is not resolvable by adding more smooth templates; it requires the admission of a qualitatively different template kind (recursive, scale-coupling templates).

This observation is more than a mathematical curiosity. It establishes that ontological deficit can be *structural* rather than merely quantitative: not a matter of having too few templates, but of having the wrong kind.

3 Admissibility and Ontological Deficit

3.1 Admissible distinctions

The Projection Theorem tells us that distinction collapse is inevitable. The theory of admissibility identifies which collapses constitute failures.

Definition 3.1 (Admissibility). Let τ be a task or purpose. A distinction $d_v \in V$ is *admissible* for τ if collapsing d_v would impair performance on τ . The *admissibility manifold* $\mathcal{A}_\tau \subseteq V$ is the set of all distinctions admissible for τ .

Before developing the consequences of admissibility, we supply the formal definition of reachability that the framework requires throughout.

Definition 3.2 (Reachability). A distinction d_v is *reachable* from a representation $\pi : \Omega \rightarrow M$ if there exists a computable function $R_v : M \rightarrow \{0, 1\}$ such that $R_v(\pi(x)) = \mathbf{1}[d_v(x, \cdot)]$ for all x in an appropriate reference class. We call R_v a *recovery operator* for d_v .

The *reachability set* of π relative to task τ is

$$\mathcal{R}(\pi, \tau) = \{d_v \in \mathcal{A}_\tau : \exists R_v \text{ as above}\}.$$

A representation *understands* domain Ω for task τ to the degree that $\mathcal{A}_\tau \subseteq \mathcal{R}(\pi, \tau)$.

Reachability, so defined, is the formal counterpart of the Witness Principle introduced in Section 8: a distinction is reachable precisely when a recovery operator witnesses it. The reachability set is therefore the set of distinctions for which constructive evidence exists.

Several features of admissibility deserve immediate emphasis.

First, admissibility is *task-relative*. The distinction between two chemical compounds that are pharmacologically identical but isotopically distinct is admissible for nuclear physics but inadmissible for most pharmacological tasks. This relativity is not a defect of the framework but a constitutive feature of understanding: understanding is always understanding *for some purpose*.

Second, the admissibility manifold has a natural stratification. Some distinctions are admissible for all tasks in a domain (structural distinctions); others are admissible only for specific interventions (functional distinctions); others only for counterfactual reasoning (causal distinctions). These strata correspond to different levels of understanding.

Third, admissibility is not observable from outside the task. A representation may appear faithful on observational tasks while collapsing distinctions required for interventional or causal tasks. This is the source of the prediction trap.

Definition 3.3 (Projection failure). A projection π *fails* relative to task τ if it collapses any distinction in $\mathcal{A}_\tau$. The *admissibility distortion* is

$$D_A(\pi, \tau) = \Pr_{(x,y) \in \mathcal{A}_\tau} [\pi(x) = \pi(y)].$$

A representation with $D_A = 0$ is *faithful* for τ ; with $D_A > 0$ it has a *projection failure*.

3.2 Ontological deficit

Ontological deficit is a stronger condition than projection failure. It arises when the representational vocabulary is incapable of expressing a distinction at all, not merely when a distinction has been collapsed for a particular input.

Definition 3.4 (Ontological deficit). A representation π has an *ontological deficit* with respect to distinction d if no state of M corresponds to drawing the distinction d : the distinction is not merely collapsed but inexpressible.

Formally, there is no partition of M that represents d , even approximately.

Example 3.5 (Pre-Darwinian natural history). Pre-Darwinian taxonomy drew distinctions between species by morphological similarity. The distinction between homologous and analogous structures — structures sharing a common ancestor versus structures convergently evolved — was not merely collapsed by pre-Darwinian taxonomy but inexpressible within it, because the vocabulary of common descent did not exist. This is an ontological deficit, not a projection failure.

Example 3.6 (Language model token prediction). A language model trained solely on next-token prediction may achieve low perplexity while having an ontological deficit with respect to the distinction between *true* and *false but fluent* continuations. The distinction is not represented in the model’s output space if no training signal distinguished it; no linear probe on the model’s internal representations will reliably recover it.

The practical significance of ontological deficit is that it is *invisible from within*. A framework with an ontological deficit cannot, using its own resources, detect the deficit. It experiences the consequences — persistent anomalies, residual structure that resists compression, asymmetric prediction errors — but cannot name their source. The deficit becomes visible only from the perspective of a successor framework that has admitted the missing distinction.

4 Residual Structure Density and the Pressure Toward Collapse

4.1 Definition and properties

Ontological deficit is a qualitative description of representational failure. The following construction provides a quantitative measure.

Definition 4.1 (Residual structure density: ideal and operational). Let $\pi : \Omega \rightarrow M$ be a representation and $\mathcal{T}$ the current template library.

The *ideal residual density* at $x \in \Omega$ is

$$\Phi^*(x) = K(x \mid \mathcal{T}) - K(x \mid \mathcal{T}^*)$$

where $K(\cdot \mid \cdot)$ is conditional Kolmogorov complexity and $\mathcal{T}^*$ is the library achieving minimal description length. Φ^* measures the theoretical gap between current and optimal understanding; it is the benchmark against which progress is assessed.

The *operational residual density* at $x \in \Omega$ is

$$\Phi_{\text{op}}(x) = \min_{t \in \mathcal{T}} d_{\mathcal{A}}(x, \mathcal{M}_t)$$

where $\mathcal{M}_t$ is the manifold generated by template t and $d_{\mathcal{A}}$ is a metric on the admissibility space. Φ_{op} is the distance from x to the nearest template boundary: it is computable, measurable, and directly actionable.

We write Φ when referring to either form. The total residual is $\bar{\Phi} = \int_{\Omega} \Phi(x) d\mu(x)$.

The two quantities serve complementary roles. Φ^* provides the conceptual foundation: it is what the framework is ultimately measuring. Φ_{op} provides the research instrument: it is what practitioners can compute as a distance from data points to the learned manifold, and its gradient $\nabla\Phi_{\text{op}}$ gives the direction in which new template admissions are most likely to be productive. This separation — between the theoretical ideal and the operational proxy — is standard practice in information-theoretic frameworks and insulates the Φ concept from the uncomputability objection while retaining its explanatory force.

Three properties hold for both forms.

First, $\Phi \geq 0$ everywhere, with equality if and only if the current vocabulary is optimal for x .

Second, Φ is *structured*: it is concentrated at points where the representation fails. Structured concentration of Φ (systematic residual rather than random noise) is the operational signature of ontological deficit. Random residuals do not define a gradient; structured ones do.

Third, Φ exerts *pressure* on the representation. The gradient $\nabla\Phi_{\text{op}}$ defines the direction in which new template admissions are likely to reduce $\bar{\Phi}$. The anomaly field is the map of the investigation frontier.

4.2 The pressure toward collapse

A template library $\mathcal{T}$ under pressure from a concentrated Φ field is in an unstable equilibrium. New templates can be admitted — speculatively, on credit — if they reduce the total description length:

$$\Lambda(\mathcal{T}) = L(\mathcal{T}) + \sum_{x \in \mathcal{O}} L(x | \mathcal{T})$$

where $L(\mathcal{T})$ is the cost of the library itself (model complexity) and $\sum L(x | \mathcal{T})$ is the total residual cost.

A template t is *admitted* if $\Lambda(\mathcal{T} \cup \{t\}) < \Lambda(\mathcal{T})$. This Minimum Description Length criterion is the formal analog of Ockham's razor: templates earn their place by explaining more than they cost.

When a successful template is admitted, the Φ field undergoes a rapid transition: regions of high residual that the new template addresses are resolved into the crystallized entropy field S . The drop in $\bar{\Phi}$ is discontinuous if the template addresses many observations simultaneously.

This discontinuity is the collapse event.

5 Paradigm Change as Distinction Reorganization

5.1 The Kuhnian framework revisited

Kuhn's account of scientific revolutions [6] identifies a pattern: long periods of normal science, punctuated by crises and revolutionary reorganizations. This account is persuasive as history but resists formalization. What exactly is a paradigm? Why do anomalies accumulate? What makes a revolution rather than a refinement?

The distinction-geometry framework provides precise answers. The translation below is not merely metaphorical; each entry on the right admits mathematical treatment.

Definition 5.1 (Kuhnian translation). Under the distinction-geometry framework:

paradigm	$\longleftrightarrow$	distinction structure $\mathcal{D} = (\Omega, \mathcal{G}_{\mathcal{D}})$
normal science	$\longleftrightarrow$	local optimization within $\mathcal{D}$
anomaly	$\longleftrightarrow$	residual structure density $\bar{\Phi}$
anomaly pressure	$\longleftrightarrow$	$\bar{\Phi}$
crisis	$\longleftrightarrow$	structured, intractable $\bar{\Phi}$
revolution	$\longleftrightarrow$	distinction reorganization (collapse event)
Kuhn loss	$\longleftrightarrow$	distinction retirement
incommensurability	$\longleftrightarrow$	non-overlapping admissibility manifolds

The translation makes Kuhnian dynamics quantitative in the following precise sense.

Proposition 5.2 (Quantitative Kuhn). Under the distinction-geometry framework:

- (i) Anomaly pressure is measured by total residual density: pressure = $\bar{\Phi}$.
- (ii) Normal science is operation in a regime where $\frac{d\bar{\Phi}}{dt} \approx 0$ under routine template additions.
- (iii) Crisis occurs when $\bar{\Phi}$ is both large and structured: residuals form a nonzero gradient $\nabla\bar{\Phi}_{\text{op}} \neq 0$.
- (iv) Revolution is a collapse event characterized by $\Delta\bar{\Phi} \ll 0$: a discontinuous drop in total residual following the admission of a new distinction d^* .

Whether or not the formalism is fully computable in historical cases, the conceptual transformation is important: paradigm change ceases to be a historical narrative and becomes a dynamical process with measurable precursors (structured $\bar{\Phi}$) and measurable outcomes ($\Delta\bar{\Phi}$). The sociology of science becomes, in principle, physics of information.

5.2 Normal science as local optimization

Within a fixed distinction structure $\mathcal{D}$, the archive performs routine encoding: new observations are matched to existing templates, residuals are minimized within the current vocabulary, and the reachability set $\mathcal{R}(\rho, \tau)$ expands incrementally. This is the

operational definition of normal science: progress that does not require new distinctions, only better use of existing ones.

Normal science is not stagnant. Within a stratum, significant progress is possible. But every stratum has a ceiling: the structural barrier defined by the distinctions it cannot express. The ceiling becomes visible when $\bar{\Phi}$ accumulates in ways that resist normal-science compression — when anomalies are not random but systematic.

Definition 5.3 (Stratum ceiling). The *ceiling* of a distinction structure $\mathcal{D}$ is the minimum description length achievable within $\mathcal{D}$:

$$\Lambda^*(\mathcal{D}) = \min_{\mathcal{T} \subseteq \mathcal{D}} \Lambda(\mathcal{T}).$$

The ceiling is positive if $\mathcal{D}$ has an ontological deficit relative to the data-generating process.

5.3 Anomaly accumulation as pressure

Anomalies in Kuhn’s account are observations that the paradigm cannot explain. In the distinction-geometry framework, they are regions of high, structured $\bar{\Phi}$: observations that resist compression under the current template library.

The crucial word is *structured*. Random residuals are uninformative. Structured residuals — residuals that have a pattern, that vary systematically with known variables, that cluster in the same regions across independent observers — carry information about the missing distinction. They define the misfit gradient: the direction in which new template admissions are likely to be productive.

Anomaly accumulation is therefore not a pathology but an information-bearing process. The archive’s persistent anomalies are its research program: they are the pressure that drives collapse.

5.4 Revolution as distinction reorganization

A scientific revolution is a collapse event at civilizational scale: a transition $\mathcal{D} \rightarrow \mathcal{D}'$ in which a new distinction d^* reorganizes the entire distinction structure, reduces $\bar{\Phi}$ discontinuously, and enables rematching of the accumulated observational archive.

Proposition 5.4 (Structure of a revolution). *A distinction reorganization event $\mathcal{D} \rightarrow \mathcal{D}'$ driven by the admission of d^* has the following structure:*

- (i) **Inexpressibility phase:** d^* is not reachable in $\mathcal{D}$; its consequences appear as structured $\bar{\Phi}$.
- (ii) **Crystallization phase:** d^* becomes expressible through new vocabulary, instrumentation, or formal notation. At this stage d^* is not yet integrated into $\mathcal{D}$.
- (iii) **Reorganization phase:** The admission of d^* retroactively restructures $\mathcal{D}$. Cases previously classified together are separated; cases previously separate are unified. $\bar{\Phi}$ drops discontinuously.

- (iv) **Rematching phase:** The observational archive is reinterpreted under $\mathcal{D}'$. Old observations yield new descriptions, often with greater precision than was possible at the time of observation.

Example 5.5 (Caloric to thermodynamic entropy). The caloric theory treated *amount of caloric* as an admissible distinction. Thermodynamics retired this distinction and admitted *entropy* (the distinction between states with many and few microstates). The reorganization collapsed the caloric distinction (reachable in the old stratum but retired in the new) while admitting the entropy distinction (inexpressible in the old stratum).

The rematching phase was extensive: steam engine efficiency data, previously described using caloric quantities, was reinterpreted in entropy terms, yielding Carnot's efficiency theorem and its consequences for what is and is not possible for heat engines. The observational archive had not changed; the distinction structure had reorganized.

5.5 Kuhn loss and distinction retirement

Kuhn loss is the observation that something may be expressible within an old paradigm that is not expressible within its successor. In the distinction-geometry framework, Kuhn loss is distinction retirement: a distinction d that was part of $\mathcal{D}$ may not appear in $\mathcal{D}'$.

This is not symmetric with distinction admission. A retired distinction was real within the old framework — it supported genuine inferences and successful manipulations — but the new framework explains those successes without it, or explains them as approximations of a more fundamental distinction.

Kuhn loss does not refute the progress account of science. It refines it: progress is the growth of the reachability set over the *admissible* distinctions, which may require retiring distinctions that were previously admitted but turn out to be approximations, confabulations, or ontological artifacts of the old vocabulary.

5.6 Incommensurability without relativism

Kuhn's incommensurability thesis has often been read as implying that paradigms cannot be compared and that scientific revolutions have no rational basis. The distinction-geometry framework dissolves this difficulty.

Principle 5.6 (Local incommensurability, global comparability). Two distinction structures $\mathcal{D}$ and $\mathcal{D}'$ are *locally incommensurable* on a distinction d if $d \in \mathcal{D}$ but drawing d is inconsistent with the organizing principles of $\mathcal{D}'$ (or vice versa).

But $\mathcal{D}$ and $\mathcal{D}'$ are *globally comparable* if their admissibility manifolds overlap on a sufficiently large common ground.

Rational evaluation across a revolution proceeds via the common ground: observations expressible in both frameworks, tasks performable under both distinction structures, predictions where the structures agree or disagree in testable ways.

Incommensurability is local; comparability is global. This is why revolutions are difficult (locally, the new vocabulary is alien) but not irrational (globally, the new structure explains more of the shared observations and reduces $\bar{\Phi}$).

6 The World-Model Crossover Theorem

6.1 The crossover condition

Every representation starts as a summary of observations. Initially, observations organize the model: each new datum updates the template library, and the model's content is determined by what it has seen. There is, however, a threshold beyond which the model's internal structure becomes more informative about the domain than any individual observation.

This is the World-Model Crossover.

Principle 6.1 (World-Model Crossover). Let $\mathcal{M}$ be a representation of domain Ω built from a sequence of observations $\{O_1, O_2, \dots, O_n\}$. Define the *crossover threshold* n^* as the smallest n such that

$$I(\mathcal{M}_n; \Omega) > \max_{i \leq n} I(O_i; \Omega)$$

where $I(\cdot; \cdot)$ denotes mutual information.

For $n < n^*$, the observations are the primary information-bearing objects about Ω ; the model is their summary.

For $n \geq n^*$, the model is the primary information-bearing object; individual observations are perturbations to the model.

An equivalent formulation in the distinction-geometry framework: n^* is the point at which $\mathcal{R}(\pi, \tau)$ becomes independent of the size and recency of the training dataset — the model's internal geometry, rather than any individual observation, provides the best guide for future inference.

Remark 6.2. The mutual-information inequality is an informative criterion but not yet a theorem with a constructive proof in the general setting. The formal crossover condition remains a conjecture in full generality. Its value is explanatory: it identifies the qualitative transition and defines what would need to be established for the claim to be proved.

The crossover is not merely quantitative. It marks a *qualitative* transition in the epistemological role of the model.

Principle 6.3 (Directionality Principle). Below n^* :

$$\text{observations} \longrightarrow \text{model} \quad (\text{observations organize theories})$$

Above n^* :

$$\text{model} \longrightarrow \text{observations} \quad (\text{theories organize observations})$$

The directionality is informational, not causal. Below the crossover, an observation is primarily *data*: something to be explained by the model. Above the crossover, an observation is primarily *evidence*: something that confirms, disconfirms, or anomalously pressures the model, adding to the Φ field if it cannot be absorbed.

The Directionality Principle is the epistemologically significant claim. The inequality

in Principle 6.1 is its formal vehicle; the principle is what readers will recognize in the actual practice of mature sciences. Newtonian mechanics, once established, did not wait for further pendulum swings to organize its claims about motion. It used its own internal structure to predict pendulum behavior, and individual pendulum swings became tests of the theory rather than its constituents. The crossover had occurred.

This transition defines the boundary between data collection and theory formation, between empirical science and theoretical science, between an archive and a world-model.

6.2 The byte as a unit of surprise

The crossover has a striking operational consequence for storage.

Below n^* , the cost of storing new observations tracks the *data rate* of the sensors: more observations, more storage.

Above n^* , the cost tracks the *novelty rate* of the world:

$$\text{storage cost} \approx H(O \mid \mathcal{M})$$

where $H(O \mid \mathcal{M})$ is the entropy of an observation *given the model*. If the model already knows what the sensors will record, the storage cost approaches zero. A 16K camera recording an empty, unchanging room costs almost nothing to store above n^* , because the model already knows the room.

The byte is no longer a unit of measurement; it is a unit of *surprise*. The archive spends bits only on what its model could not predict.

6.3 Crossover and the diversity requirement

The crossover threshold n^* is not reached by accumulating more observations of the same type. It requires *diversity of kernels*: observations from different measurement modalities that triangulate the same underlying domain from different angles.

Proposition 6.4 (Diversity requirement). *Let $\{K_1, K_2, \dots, K_m\}$ be measurement kernels (e.g., visual, acoustic, geometric). The joint information $I(\{K_i\}; \Omega)$ exceeds $\max_i I(K_i; \Omega)$ when the kernels expose orthogonal aspects of the domain.*

Formally, n^ is minimized by maximizing the diversity of the kernel family:*

$$n^* \propto \frac{1}{\sum_{i \neq j} I(K_i; K_j \mid \Omega)}$$

Highly correlated kernels (many cameras from similar angles) delay the crossover; orthogonal kernels (visual plus acoustic plus structural) accelerate it.

No single modality reaches the crossover alone. Genuine world-models emerge from the integration of multiple measurement kinds, because no single kind exposes the full structure of the domain.

This is not merely a technical observation. It explains why scientific understanding depends on the coordination of multiple experimental traditions, and why disciplines that share data across modalities progress faster than those that do not.

6.4 Post-crossover rematching

Above the crossover, a new observation that conflicts with the model does not merely update the model's beliefs. It generates an anomaly in the Φ field, creating pressure for a reorganization.

The post-crossover archive therefore exhibits the same dynamics as the pre-revolution scientific community in Kuhn's account: anomalies accumulate, misfit gradients point toward missing distinctions, and collapse events are triggered by template admissions that resolve the pressure.

The crossover is not a permanent state but a moving threshold. Each collapse event raises the model's informational content, but also opens new strata of the domain that the model cannot yet address. The Φ field never reaches zero; it reorganizes. Understanding is not a destination but a process of continual reorganization.

7 Memory as Distinction Reorganization

7.1 The recording illusion

The folk model of memory presents recollection as retrieval: a record stored at encoding is located and returned intact. This picture is not merely imprecise but structurally wrong in a way that the distinction-geometry framework clarifies.

A representation stores not observations but *the distinctions an observation supported*. Recollection is reconstruction from those distinctions as they are organized in the current distinction structure. Because the current structure differs from the structure at encoding, recollection is necessarily reconstruction, not retrieval.

Principle 7.1 (Memory as reconstruction). Recollection of event e is the inference to the best distinction-preserving completion of the current distinction structure $\mathcal{D}_t$ consistent with the partial record encoded at time of observation.

Since $\mathcal{D}_t \neq \mathcal{D}_{t_0}$ in general, the recollection is filtered through distinctions that did not exist at encoding. This is not a defect but a constitutive feature: the past is rendered in the vocabulary of the present.

The phrase "rendered in the vocabulary of the present" requires a guard against a common misreading.

Principle 7.2 (Representational Revision Principle). Distinction reorganization modifies the *representation* of prior observations, not the observations themselves.

Formally: a collapse event $\mathcal{D} \rightarrow \mathcal{D}'$ changes the encoding $\pi_{t'}(e) \neq \pi_t(e)$ for observations e encoded before the event, but the observation $e \in \Omega$ itself is unchanged.

The past is not rewritten. The representation of the past is rewritten.

This principle insulates the framework from the objection that it conflates epistemology with ontology — an objection that has been leveled against strong readings of the free-energy principle and related Markov blanket formalisms. The distinction-geometry framework is entirely epistemological: it concerns how observations are encoded and recovered, not what the observations themselves are. The revision of a memory under reconsolidation changes the *render*, not the event.

This prediction is well confirmed. Bartlett’s experiments on reconstructive memory [1], the misinformation effect studied by Loftus [8], and the reconsolidation research of Nader and colleagues [9] all document exactly the behavior the framework predicts: recall is shaped by post-encoding knowledge, schema, and belief.

7.2 Reconsolidation as local collapse

Memory reconsolidation — the phenomenon in which retrieved memories become temporarily labile and subject to modification — is, in the distinction-geometry framework, a local collapse event.

When a memory is retrieved, it is reconstructed within the current distinction structure $\mathcal{D}_t$. If $\mathcal{D}_t$ contains distinctions that were absent at encoding time t_0 , the reconstruction may differ from the original. The reconsolidation window is the period during which the reconstructed memory is re-encoded using the current (potentially reorganized) vocabulary.

This is not distortion in any pejorative sense. It is the faithful application of the distinction-reorganization process to the domain of episodic memory. The memory becomes more *reachable* — more efficiently represented within the current distinction structure — at the cost of some fidelity to the original encoding.

Reconsolidation is therefore not a vulnerability of the memory system but evidence of its design: a rematching archive does not store frozen records but living theories, and living theories must be updated when the vocabulary improves.

7.3 The compressibility of the past

A counterintuitive prediction of the framework is that the past *becomes more compressible over time* as the distinction structure matures.

Early in learning, autobiographical memories are encoded in coarse-grained vocabularies: the memory of a trip to Paris at age twelve is encoded in whatever distinction structure was available at age twelve. As the distinction structure matures — as new templates are admitted, as more subtle distinctions become reachable — the same event can be re-encoded more efficiently. Details that were stored as raw impressions become pointers to general templates; patterns that were opaque become recognizable instances of known categories.

This is what we experience as the wisdom of retrospection: the sense that past experiences become more *intelligible* as we mature, not because the events changed but because our distinction structure has reorganized to make them more compressible.

8 Interpretability as Distinction Verification

8.1 The interpretability reframing

Mechanistic interpretability in AI research is commonly framed as the project of understanding what is happening inside a trained model: identifying circuits, features, attention patterns, and internal representations that explain the model’s behavior.

The distinction-geometry framework reframes this project. Interpretability is not primarily about *describing* internal structure. It is about *verifying* whether the model has preserved the admissible distinctions required for navigation, explanation, and intervention.

A description of internal structure that does not answer this question is a description, not an explanation.

Principle 8.1 (Interpretability as distinction verification). A mechanistic interpretability claim is *explanatory* only if it demonstrates that the identified internal structure preserves a specific admissible distinction.

An interpretability claim that identifies a feature without showing how that feature preserves a distinction in $\mathcal{A}_\tau$ for some task τ is a description of the representation, not an explanation of the understanding.

This principle has sharp practical consequences.

8.2 Chain-of-thought and the trajectory question

Large language models prompted to produce chain-of-thought reasoning exhibit markedly improved performance on tasks requiring multi-step inference. The standard interpretation is that chain-of-thought forces the model to externalize its reasoning trajectory, making previously internal steps visible and correctable.

The distinction-geometry framework raises a more pointed question:

Is the model *performing* the reasoning trajectory, or merely *describing* a trajectory it did not use?

These are not the same. A model could produce a fluent verbal description of a reasoning process that is entirely post-hoc confabulation, while the actual computation that produced the answer preserved no admissible distinctions from the described steps.

The test is not whether the chain-of-thought is coherent but whether the distinctions drawn in the chain-of-thought are preserved in the computation. If removing a step from the chain changes the model’s output in ways that track the causal role of that step, the chain-of-thought preserves admissible distinctions. If the chain is rearranged or corrupted without affecting the output, it does not.

8.3 The recoverability requirement

A stronger condition than distinction preservation is *distinction recoverability*: the ability to extract a specific distinction from the representation using a designated recovery

operator.

Definition 8.2 (Recoverability). A distinction d_v is *recoverable* from representation π via operator R_v if there exists a map $R_v : M \rightarrow \{0, 1\}$ such that $R_v(\pi(x)) = \mathbf{1}[d_v(x, \cdot)]$ for all x in the relevant domain.

Principle 8.3 (Witness principle). A claim that distinction d_v is represented in a model M requires a recovery operator R_v as constructive evidence. Without a recovery operator, the claim is an assertion, not a proof.

The witness principle is deliberately stringent. It excludes post-hoc attribution stories that are consistent with the model’s behavior but do not reflect a genuine distinction in the model’s internal structure.

This is not a demand for perfect recovery. Approximate recoverability — with known error bounds — is sufficient. But the recovery operator must exist and be demonstrated, not merely hypothesized.

8.4 Observational and interventional separation

The admissibility framework distinguishes two levels of distinction: those separating cases that look different (*observational*) and those separating cases that respond differently under intervention (*interventional*). A representation that is faithful at the observational level may fail at the interventional level.

Theorem 8.4 (Observational–Interventional Separation). Let $\pi : \Omega \rightarrow M$ be a representation and let τ_{obs}, τ_{int} denote the observational and interventional task families. Then:

$$D_A(\pi, \tau_{obs}) = 0 \not\Rightarrow D_A(\pi, \tau_{int}) = 0.$$

Observational faithfulness does not entail interventional faithfulness. Equivalently, there exist cases $x, y \in \Omega$ such that $\pi(x) = \pi(y)$ (observationally identical) but $do(x) \neq do(y)$ (interventionally distinct), where $do(\cdot)$ denotes the response to a perturbation.

Proof. It suffices to exhibit a representation that collapses an interventional distinction while preserving all observational ones. Let $\Omega = \{x_1, x_2, x_3\}$ where x_1, x_2 have identical observable properties but distinct responses to perturbation p : $do_p(x_1) = r_1 \neq r_2 = do_p(x_2)$, and x_3 is observationally distinct. Define $\pi(x_1) = \pi(x_2) = m_1$, $\pi(x_3) = m_2$. Then $D_A(\pi, \tau_{obs}) = 0$ (only observationally distinct cases are separated) but $D_A(\pi, \tau_{int}) > 0$ (the interventional distinction $d(x_1, x_2)$ is collapsed). $\square$

The theorem is a formal statement of a familiar intuition in the causal inference literature [10]: correlation does not imply intervention-reliability. Its significance here is that it gives the interpretability reframing a formal result to anchor on. A model that achieves low test error may be intervening on a statistical proxy rather than the genuine causal variable; the Observational–Interventional Separation Theorem says this is not merely possible but structurally expected when the training signal is purely observational.

The practical test is: if the model's claimed internal distinction d_v is manipulated directly (via causal intervention on the representation), does the output change in ways that track the real-world causal role of d_v ? If not, the distinction has not been recovered; it has been described.

9 Unification: Six Programs, One Geometry

The framework developed in this paper was not designed to unify independent research programs. The unification is a consequence of the framework's generality. Nevertheless, it is worth making the connections explicit.

Frozen Processes. The Frozen Processes program [18] studies what happens when a distinction structure is assumed fixed: $\mathcal{A}_t = \mathcal{A}_0$. In the language of the present paper, a frozen-process assumption blocks distinction reorganization. The consequences are exactly what Frozen Processes documents: the accumulation of Φ in the form of unexplained anomalies, the inability to rematch historical observations, and the eventual crisis that forces a reorganization. Frozen Processes is the study of the *obstruction to collapse events*.

Persistent Anomalies. The Persistent Anomalies program studies structured residuals that resist compression under the current template library. In the present framework, a persistent anomaly is a concentrated, structured region of the Φ field whose misfit gradient points toward a specific missing distinction. Persistent Anomalies is the study of *Φ dynamics and the geometry of the investigation frontier*.

Waves of Collapse. The Waves of Collapse program [16] studies the discontinuous structure of archive compression: long plateaus of normal encoding punctuated by collapse events. In the present framework, this is the Devil's Staircase of the total description length Λ under sequential template admissions. Waves of Collapse is the study of *collapse event dynamics and the Devil's Staircase structure of $\Lambda(t)$* .

Admissibility Geometry. The Admissibility Geometry program formalizes the admissibility manifold and the geometry of projection failure. It is the foundational mathematical framework from which the present paper derives its formal content. Admissibility Geometry is the study of *the structure of $\mathcal{A}_\tau$ and the metrics on the space of distinctions*.

Distinguishability Geometry. The Distinguishability Geometry program studies the metric structure of the space of distinctions: how far apart are two cases, how much information separates them, how the geometry changes under different admissibility assumptions. In the present framework, distinguishability geometry provides the metric on the distinction space within which collapse events are geodesic reorganizations. Distinguishability Geometry is the study of *the Riemannian structure of the admissibility manifold*.

RSVP (Relativistic Scalar-Vector Plenum). The RSVP program provides the field-theoretic dynamics: the equations governing how Φ , v (rematching flow), and S (crys-

tallized entropy) evolve, interact, and generate shock fronts at collapse events. RSVP is the study of *the thermodynamics of distinction reorganization*.

The six programs are cross-sections of the same geometric object: the distinction structure of a domain under pressure from accumulated residual and periodic reorganization by collapse events. Each program captures a different aspect — the obstruction to collapse, the dynamics of anomaly accumulation, the structure of discontinuous events, the geometry of admissibility, the metric on distinctions, the thermodynamic field theory of reorganization — but each is addressing the same underlying phenomenon.

10 Conclusion: The Master Claim and Its Corollaries

The argument of this paper can be compressed into seven statements.

1. **Projection destroys distinctions.** Every nontrivial representation collapses at least one distinction (Theorem 2.6). Epistemology is the theory of which collapses matter.
2. **Admissibility identifies which collapses matter.** A distinction is admissible for a task if collapsing it would impair navigation, intervention, or explanation. Projection failure is admissibility-relative: a harmless collapse for one task may be catastrophic for another.
3. **Ontological deficit is invisible from within.** A distinction that is inexpressible in the current vocabulary manifests as structured residual $\bar{\Phi}$, not as an identifiable gap. The deficit becomes visible only from the successor vocabulary.
4. **Residual density $\bar{\Phi}$ is the pressure toward reorganization.** Concentrated, structured $\bar{\Phi}$ defines the investigation frontier. Anomalies are not noise; they are the fuel for collapse events.
5. **Understanding advances through collapse events.** A collapse event admits a new distinction d^* that reorganizes the existing distinction structure, reduces $\bar{\Phi}$ discontinuously, and enables rematching of the accumulated observational archive. Scientific revolutions, memory reconsolidation, archive rematching, and ontology repair in AI are all collapse events.
6. **The World-Model Crossover and Directionality Principle mark the boundary between observation-centered and model-centered epistemology.** Below n^* , observations organize theories. Above n^* , theories organize observations. The byte becomes a unit of surprise rather than a unit of size. This is a principle and research program, not yet a theorem with a constructive proof in the general case.
7. **Interpretability is distinction verification.** A model understands a domain to the degree that it preserves and makes recoverable the admissible distinctions. Chain-of-thought, mechanistic features, and internal representations are explanatory only insofar as they recover specific distinctions via recoverable operators.

These seven statements converge on a single master claim:

Understanding is not the accumulation of descriptions but the reorganization of the distinctions from which descriptions are constructed.

And because reorganization is irreducibly historical — because it occurs in time, requires a prior state to reorganize, and produces a successor state that cannot be reached by accumulation — understanding is more precisely defined as a *trajectory*:

Understanding is a trajectory of distinction reorganization: $\mathcal{D}_0 \rightarrow \mathcal{D}_1 \rightarrow \mathcal{D}_2 \rightarrow \dots$, in which each transition expands reachability, reduces residual density, and retroactively remaps the accumulated observational archive.

This formulation resolves the tension between the static picture (understanding as a property of a representation) and the dynamic picture (understanding as a process). Both are correct at different levels of description. The static picture characterizes a given distinction structure $\mathcal{D}_t$ at a moment. The dynamic picture characterizes the sequence and its properties. The static picture is what a representation has; the dynamic picture is what an understander does.

Prediction is not understanding because a predictor can collapse admissible distinctions while achieving high accuracy. Compression is not understanding because a compressor can reduce size while discarding the distinctions required for intervention. Description is not understanding because a description can be accurate and inaccessible, faithful to the observation and unfaithful to the domain.

Understanding is something more specific: a representation that preserves the admissible distinctions of a domain, organizes them into a structure with maximum reachability, and is capable of reorganizing that structure when the residual field demands it.

The geometry of understanding is the geometry of this process. It is not the geometry of facts in a space but the geometry of distinctions in transformation.

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